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Foam EN Presentation NA

Foam Reduction in Bloomsburg

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Current Situation

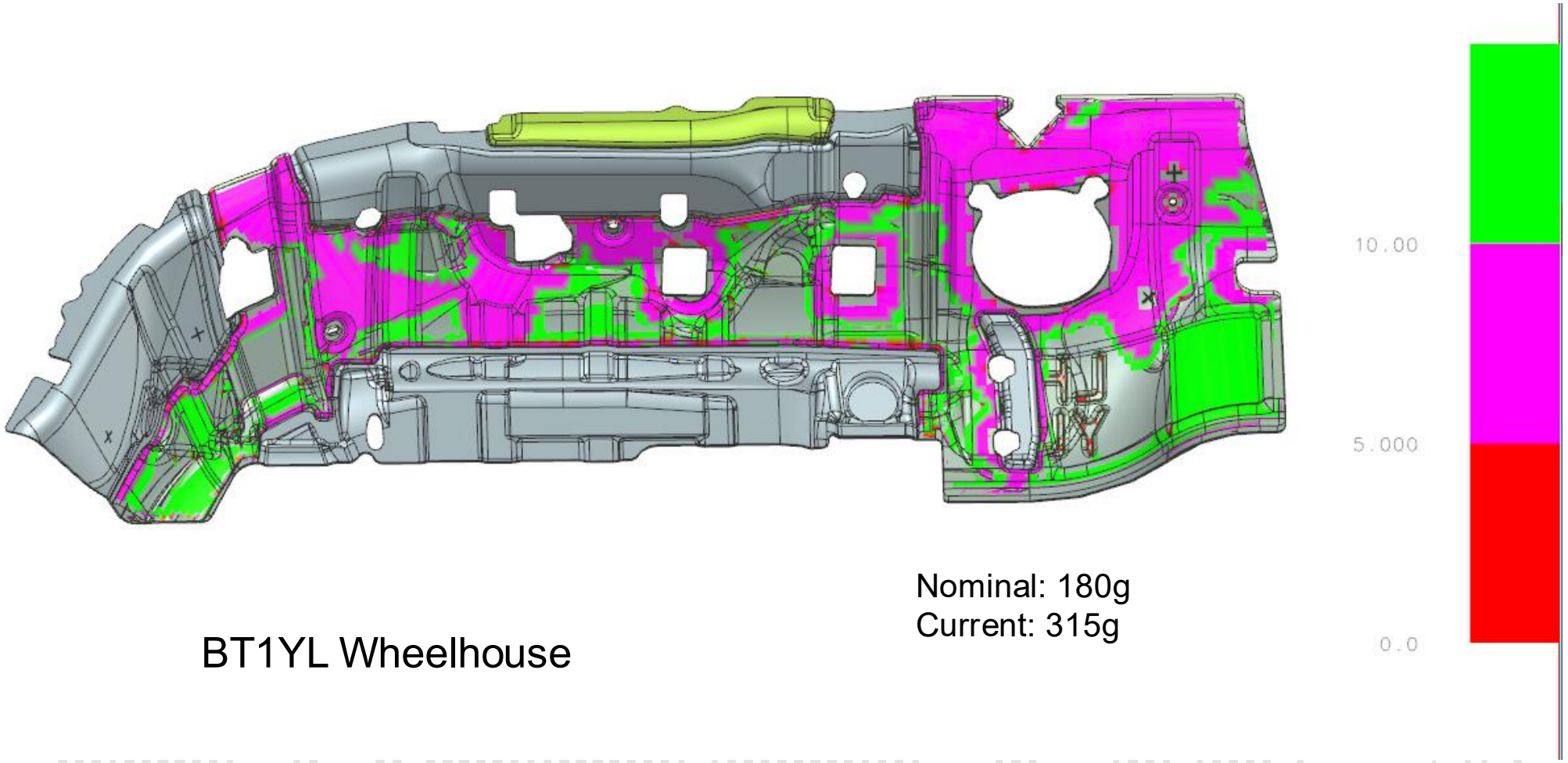
Overpack

- Bloomsburg currently has high overpack on some parts, due to very thin geometries
  - 25 – 50%
  - Geometries are overall low foam weight, but this still cuts into business case

- During production, after the foam is poured into the open tool
  - Foam rises too quickly
  - Tool closes onto the foam, collapsing it, and the foam loses its energy to fill the tool

Current Situation

Example Geometry



BT1YL Wheelhouse

Nominal: 180g
Current: 315g

Current Foam Capability
Analysis

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- Bloomsburg team measured the 31950 capability to fill flat sample tool at different thicknesses. Nominal is 50kg density.

Minium Amount of Foam Used to Fill a 1400mm x 1200mm Area						
	High Resistance		Visco Open Pour		Visco Closed Pour	
Shim	Foam	Density	Foam	Density	Foam	Density
mm	grams	kg/m^3	grams	kg/m^3	grams	kg/m^3
5	1080	128.57	976	116.19		0.00
10	1292	76.90	1226	72.98		0.00
15	1555	61.71	1534	60.87		0.00
20	1821	54.20	1850	55.06		0.00
30	2365	46.92	2632	52.22		0.00
40	3005	44.72	3378	50.27		0.00
50	3530	42.02	4194	49.93		0.00

Potential Solutions

Delayed Catalyst

- Huntsman tried modifying the current 31950 formula
  - Slow catalyst up front, to reduce rise before tool closes
  - Keep overall cure and blow properties the same, as not to affect cure time

- Huntsman was unable to finalize a working formula, so no trial was done

Potential Solutions

Low Density Visco

- Huntsman has offered an RG50 visco foam formula
  - On flat samples, this proved more capable at filling thin geometries
  - But is more expensive
  - Flat sample results warranted a pare trial

Potential Solutions

Low Density Visco

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- This table shows trial results for LDV vs current foam weights

	HR Foam Base Line Min Fill		Trial Visco Min Fill		Difference	
	Foam	Cure Time	Foam	Cure Time	Foam	Cure Time
U71x SWB	1551	90	1332	150	219	-60
U71x LWB	1850	90	1637	150	213	-60
P702	841	90	733	150	108	-60

Foam is in grams Cure Time is in seconds

	Price HR Foam	Price Trial Foam	Difference
U71x SWB	$335,481	$308,624	$26,857
U71x LWB	$182,928	$173,391	$9,537
P702	$1,689,149	$1,577,050	$112,099
			$148,493

HR Price	Trial Foam Price
3.09	3.31

Potential Solutions

Low Density Visco

- In the end, foam weight savings were achieved
  - But these savings were heavily offset by the price increase
  - There were still cost savings achieved

- Increase in cure time by 60s killed any potential savings of the project
  - Huntsman has stated that adding more catalyst to increase cure time, will also increase price

- LDV project potentially to be cancelled, due to offset of price increase vs weight savings

Potential Solutions

Aiken Visco 32085

- The visco in Aiken has proven more capable of filling thin geometries
  - But the nominal density is 70kg/m^3

- If customer can be convinced to accept RG70 density, this could be a way forward
  - Customer is already receiving 60 – 100kg density, no difference would be felt on the part

CapEx

Bulk Tank Mixer

- Bulk tank in Bloomsburg doesn’t have a mixer
  - New foam formulas will require mixing

- 100k USD to install new mixing blade
  - This will also need to be overcome by the savings

CapEx

Bubble Mixer

- Pulsair is a supplier that makes bubbler mixers
  - They quoted 12,650 USD to install 4 bubblers and a control unit to the bulk tank

- Concern would be to aerate the poly
  - To offset this, a control unit would be used, that can control the frequency and length that the bubblers activate

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